Sumit Chhajed

CONTACT

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EDUCATION

Thadomal Shahani Engineering College | 2023-2027

Bachelor of Engineering in Electronics and Telecommunication (Finished 5th semester)

- SGPA Current Semester (5): **9.14**
- CGPA Overall : **8.20**
- **Relevant Coursework : Data Structures and Algorithms (Third Year)**
- **Upcoming Coursework : Database Management System (SQL)**

Nahata College | 2021-2023

XII Grade HSC(65.67%)

Tapti Public School | 2021

X Grade CBSE (77.83%)

WORK EXPERIENCE

Python and SQL Data Analyst Intern | Dec 2025 - Jan 2026

Webgile Solutions, Nagpur, India

- Performed end-to-end data analysis using Python (pandas, NumPy) and SQL, including data extraction, cleaning, and transformation to derive meaningful insights.
- Developed and executed complex SQL queries (joins, aggregations, subqueries) to analyze large datasets and support data-driven decision making.
- Created visualizations and analytical reports using matplotlib and Excel to identify trends, patterns, and business insights.
- Applied data preprocessing and basic statistical techniques to improve data quality and ensure accurate analysis.

ACHIEVEMENTS (TECHNICAL)

2nd Runner Up at Tech-a-thon Hackathon

- 3rd Place finish in Hackathon with 90+ participating teams

Qualified SIH Internal Hackathon
2nd Runner Up at Hardware Project Exhibition TSEC 2025

CLG - COMMITTEE EXPERIENCE

Technical Head @ ISTE TSEC | 2025-26

- Led and managed technical events, workshops, and team activities
- Organized sessions and collaborated for hackathons and guest lectures
- Oversaw technical execution, planning, and problem-solving

PROJECTS

IOT Based Smart Helmet to enhance delivery workforce safety

- Developed a smart helmet to enhance rider safety using **Raspberry Pi 4 and ESP32**.
- System verifies helmet usage via an HC-SR04 ultrasonic sensor, **detects alcohol** with an MQ-3 sensor, and
- **Monitors drowsiness through eye pattern movement using a camera. Libraries : MediaPipe, OpenCV, used a pretrained ML dataset**
- **Fall detection is enabled** using an MPU6050 accelerometer.
- Safety data is sent to Firebase Realtime Database and used to control an ESP32-based servo lock.
- **The bike unlocks only when the rider is wearing the helmet, is sober, and not drowsy**. Additionally, the system uses Twilio API to send emergency SMS alerts with GPS location in the event of a fall.

DeepHider: High-Fidelity Steganography & Cryptography Engine

- Technologies: Python, OpenCV, NumPy, Hashlib, Tkinter/KivyMD
- Engineered a custom 16-bit byte-slicing algorithm to losslessly embed high-resolution image payloads into container images without visual degradation.
- Implemented a secure XOR stream cipher utilizing SHA-256 key expansion to password-protect hidden data.
- Optimized image processing speed by vectorizing pixel manipulation matrices using NumPy, eliminating slow iterative loops.
- Developed a responsive GUI featuring batch-processing capabilities and packaged the tool as a standalone cross-platform executable.

Microgrid Management System | Link

- Developed an IoT-based monitoring system for solar or wind microgrids in rural areas, providing real-time data on energy generation, storage, and consumption
- A hardware-software prototype **improved sample microgrid efficiency by 15%**, with a local mobile app for community operators to manage energy distribution

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SKILLS

Languages - Python, C++, JavaScript, SQL, HTML, CSS
Machine Learning : Scikit-learn, Pandas, Model Training,OpenCv
Tools : Visual Studio, VS Code, Git, GitHub, Android Studio
Core Skills: Basic Linux commands, Data Structures, OOP.
Databases: SQL Server, MySQL, Functions, Query Optimization
Hardware Design tools (KiCad, LTSpice, Fritzing)
Cloud Platforms (AWS, Google Cloud - Firebase)
Embedded Systems (STM32, ESP32, RaspberryPI, other microcontrollers)
